



Radiated Tire Noise Source Level Evaluation Procedure

1 Scope

Note: Nothing in this standard supercedes applicable laws and regulations.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. The purpose of this test procedure is to measure the radiated tire noise source level under steady-state operating conditions.

1.2 Foreword. Overall interior vehicle noise comes from three primary sources:

- Powertrain noise
- Windnoise
- Tire noise

This procedure describes a method for measuring tire noise away from other sources. Tire noise is measured in the vicinity of the tire contact patch. The source level is, therefore, measured to address only the airborne noise portion which passes around the vehicle exterior and further penetrates into the vehicle.

The measurements can be made either on a test trailer or a vehicle. The test on a trailer has an advantage that the measurement will be under the same load and independent of the vehicle condition such as toe-in angle. However, the test on a given vehicle shows the result that corresponds to a real tire/road interaction. The latter test is easy to operate, but its result is dependent on the vehicle suspension system. If there is no trailer, it is recommended to always use the same vehicle such as a pickup or van to do the same tests and to always measure the tire noise from the rear tires and set neutral toe-in if possible.

The road tests will be performed mainly on a smooth asphalt road as indicated in GMW8460 or ISO 10844. The tire noise tests can be done either on road or on dynamometer (dyno). For the latter noise test, the dyno shell surface profile should be comparable to the VTS road surface. The use of the dyno gives all weather or all season capability and improves data repeatability and work efficiency. Appendix A describes the alternative on-dyno method.

1.3 Applicability. This test procedure applies to passenger car tires and light truck tires.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

ANSI 12.35-1990 ISO 10844
ASTM F2493

2.2 GM Standards/Specifications.

GMW8460

2.3 Additional References.

Acoust, J. (1980). Transfer Function Method of Measuring Acoustic Intensity in a Duct System with Flow, *Soc. Am* 68 (6), Dec.

3 Resources

3.1 Facilities. Smooth asphalt road per GMW8460 or ISO 10844. The test may be carried out on dyno road shells as long as they are comparable to the VTS specified roads. The 80 km/h smooth asphalt road VTS condition should be used to set the radiated tire noise specification requirement. This condition can best accentuate airborne tire noise. Other roads and speed condition can be used for other tire noise measurement purposes.

3.2 Equipment. The following equipment and instrumentation, or equivalent, is required to perform this test. See Figures D1 through D3 for the photos of a microphone pair setup and its fixtures on a tire noise trailer or a vehicle.

3.2.1 One half inch microphones and corresponding preamplifiers are required. Two (2) phase-matched pressure microphones are preferred.

3.2.2 A signal analyzer or data acquisition system is required to conduct the cross power measurement between two sound pressure signals in a frequency range spanning 0 to 8192 Hz.

3.2.3 High-pass filters are required to filter out low frequency turbulence induced noise to improve the signal-to-noise ratio of the measurement. If the measurement system has 24 bit A to D conversion, the user does not necessarily need to install high-

pass filters. Ensure that the filters will not cause any phase distortion between two microphone signals.

3.2.4 Test trailer or vehicle with microphone boom and wheel adapters for appropriate wheel bolt patterns (see Figures D1 to D3).

3.2.5 If required for towing, a truck with quiet tires must be of sufficient engine power to tow the trailer.

3.3 Test Vehicle/Test Piece. Two (2) identical tires (or two different tires with equivalent diameters) are requested per individual test with one tire on each side of a trailer or a vehicle. Only one tire is tested and, shall be suitably identified before the test.

3.4 Test Time.

Calendar time:

1 to 2 tires: 1 to 2 days

3 to 8 tires: 2 to 5 days, depending on weather conditions, tire numbers, tire size difference, whether Standard Reference Test Tire (SRTT) (SRTT, recommended to be ASTM F2493 16 in. Tire) tests are included or not, and run numbers for each condition, etc.

Test hours:

Four (4) to 32 h, depending on sample size, adapters, and load adding, etc.

One (1) to 2 h (post processing/managing database)

One (1) to 2 h (equipment setup)

Note that the test time estimated here requires two runs for each condition to maintain data repeatability.

3.5 Test Required Information.

- Tire manufacturer
- Tire trade name
- Tire size
- Tire load/speed index
- GM tire type (such as ALS, AL2, AL3, HW4, AT, and RFT, etc.)
- GM construction number
- Inflation pressure
- Test conditions
- GM vehicle program name or competitor's vehicle names (if their tires are tested), etc., shall be given for database management or cross-reference. In particular, the construction number of a tested tire shall be provided if the tire is not a production tire.

3.6 Personnel/Skills. To execute this test, at least one technician or one engineer skilled in noise and vibration data acquisition, analysis, equipment usage, and test setup is required.

4 Procedure

4.1 Preparation.

4.1.1 Obtain wheels of an appropriate size, mount the tires and balance the tire/wheel assemblies dynamically.

4.1.2 Install appropriate or universal hub adaptors if using the trailer or a vehicle in tests.

4.1.3 Mount the tire/wheel assemblies. Use tires per 3.3 to maintain the trailer or vehicle horizontally level or the same height from both tires of the trailer or vehicle. The difference shall be less than 3% of the height difference, between a left tire and a right tire, divided by the span distance between these two tires.

4.1.4 Apply a standard vertical test load to each tire. The standard load for tire testing is 70% of the Tire & Rim Association (T&RA) maximum load or 70% of The European Tyre and Rim Technical Organization (ETRTO) standard load capacity stated on the tire. The load value should be rounded up to the nearest 50 kg to simplify measurement process. If the estimated load is more than 750 kg, use 750 kg for the standard load.

4.1.5 Adjust the cold inflation pressure to the given vehicle specifications; otherwise to 207 kPa (or 30 psi) for passenger vehicle tires.

4.1.6 Set toe-in and camber angles for both tires to zero degrees or as small as possible. If using a trailer, adjust the angle to zero under the standard load of 550 kg.

4.1.7 Tire Release Lubricants or Other Chemicals. If the surface of brand new tires has residue of mold release lubricants or other chemicals such as glue, the operators shall clean them with Adhesive Cleaner (containing Xylene) or other Xylene equivalents (which will not damage the rubber surface) and paper towels before running tire break in. Follow correct safety and ventilation procedures when using chemicals.

4.1.8 Tire Break-In. For brand new tires, it is recommended to run them at least for one half hour at a speed of 80 km/h or higher on road to eliminate or reduce the surface stress unevenness and to allow the other tire components to begin working together. **Break-in** can be on any surfaces except for the severe coarse road surface.

4.1.9 Tire Wear. The tire noise test requests the tire to be new and free of any noticeable wear. New tires can be identified when their mold tread vents are clearly visible or not worn off. If new tires are submitted for multiple validation tests, the noise test shall be requested ahead of the vehicle dynamics tests (such as handling and traction) which may cause tire surface wear. Avoid tire wear caused by aggressive driving. If the tires are dismounted from a vehicle, it is recommended to use the rear tires which less wore off.

4.1.10 Tire Warm Up. Run test tires more than 8 km or 6 minutes at 80 km/h prior to testing in order to bring the tires up to their working temperature.

4.1.11 Remove small stones from tread grooves prior to test runs.

4.1.12 Ensure the speedometer of the tow vehicle is accurate.

4.2 Conditions.

4.2.1 Environmental Conditions. See GMW8460.

4.2.1.1 Wind speed shall be less than 12 km/h.

4.2.1.2 Ambient temperature shall be between 5 °C (41 °F) and 35 °C (95 °F).

4.2.2 Test Conditions. Deviations from the requirements of this standard shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.3 Instructions.

4.3.1 Equipment Installation.

Note: Do not set the microphones in position prior to driving to the beginning of the test road to avoid damage from access roads, curbs, etc.

4.3.1.1 The microphone amplitude shall be calibrated and the microphone pair shall be phase-matched within 0.5 degrees up to 6400 Hz using a signal analyzer or data acquisition system. In order to evaluate phase matching, the phase of the cross-power of the microphone pair when subject to the same random noise in the upper chambers of B&K Type 3541 Intensity Calibrator (or equivalent) should be measured. If the phase is not less than 0.5 degrees, they shall be replaced by a new pair that meets the 0.5 degree phase matching requirement. The operator can also use the phase-matched microphones.

4.3.1.2 Install microphone preamplifiers into the microphone preamplifier spacer/holder. (See Figure D1 for the photo of a microphone pair.) Make certain that they are installed to the same depth in the holder. By convention, Channel

number one of the pair is assigned to the microphone closer to the test tire.

4.3.1.3 Microphones are spaced at 16 mm for the sound intensity measurement.

4.3.1.4 Prior to tests, microphone turbulence wind screens (trimmed to fit the two microphones) shall be installed to reduce low-frequency turbulence/noise generated by the rotating tire and wind.

4.3.1.5 Mount the preamplifier spacer/holder on the intensity fixture which is attached to the test trailer or a vehicle as shown in Figures D1 through D3.

4.3.1.6 Turn on instrumentation and allow five minutes to warm up.

4.3.1.7 Set "High Pass" filters at 250 Hz, 0 db gain.

4.3.2 Test and Tire Documentation. Record all the factors related to the test tire as well as all ambient factors. Factors related to tires are:

Example:

- Manufacturer/supplier: Goodyear or GDY
- Model name: Eagle GA
- Size: P215/65R15
- Load and Speed Index: 98H
- GM Tire Type: ALS/AL2/AL3/HW4
- Construction number: 8X1040B
- Load: 550 kg
- Inflation pressure: 207 kPa or 30 psi
- Ambient factors to be recorded are:
 - Temperature: 30 °C
 - Wind speed: 4 km/h
 - Wind direction: East

4.3.3 Operation Details.

4.3.3.1 Attach 12.7 mm (½ in) wind screens onto the microphone heads.

4.3.3.2 Use the standard test load (see 4.1.4).

4.3.3.3 Immediately prior to testing, set the microphone probe in the measurement position at the noise test road: Lower the microphone pair and adjust it so that the pair centerline is 70 mm above the road surface, 100 mm outboard away from the sidewall (the most stretching-out point of the convex shape) of the test tire, with the microphone diaphragms in line with the leading edge of the tire contact patch as shown in Figure D1 or Figure D4.

Because the tire patch is usually in a convex shape, the exact leading edge and trailing edge shall be measured at the centerline of the tire (perpendicular to the axle, centerline of tire contact patch). An acceptable practice of determining the leading edge line or trailing edge line is to use a

paper sheet that slides in between tire and ground unit it touches the centerline.

4.3.3.4 Monitor the noise levels through headphones during the entire recording period in order to identify any occurrence of unusual sounds. Isolate the occurred unusual sound. Check the signal analyzer or data acquisition system for overloads.

4.3.3.5 Measure the cross-power spectra at given conditions, usually at 80 km/h on smooth asphalt and at 60 km/h on stud damaged concrete. At least 8 s of data to average the cross-power spectra are required.

Note: For consistency, use the same road section and the same speed in tests.

4.3.3.6 Move the microphone pair so that their diaphragms are in line with the trailing edge of the tire contact patch.

4.3.3.7 Repeat the step in 4.3.3.5 to obtain the cross-power spectrum at the trailing edge. The procedure also allows two pairs of microphones mounted in a test fixture to measure both the leading and trailing the cross-power spectra simultaneously.

4.3.4 Data Analysis. See Appendices B and C.

4.3.5 Data Validity. Check two runs for the same conditions to see whether both curves are within 1 dB. Also check the validity of results by reviewing SRTT spectra to determine whether they fall within an acceptable range (see 4.3.5.2) as well as by reviewing the coherence, direction, and pressure-to-intensity index. Reject test data if they do not meet requirements.

4.3.5.1 If the following requirements are not met, tests shall be rerun or data in the affected frequency range shall be excluded.

4.3.5.1.1 Direction. The negative intensity shall not occur above 448 Hz and below 4000 Hz.

4.3.5.1.2 Pressure-to-Intensity Index. This value shall be less than 10 dB for valid data.

4.3.5.1.3 Coherence: Values shall be between 0.5 and 1.0. In frequency bands where data are less than 0.5, the data shall not be used since it is most likely contaminated by wind gusting or non-tire related noises.

4.3.5.2 Use the SRTT results to verify whether the overall system is operating correctly or not. Usually measure the SRTT results at the start of each batch test. Spectra are plotted to compare the SRTT previous results and to examine the data consistency and repeatability.

5 Data

5.1 Calculations. The intensity spectrum calculation is based on two-microphone cross-spectral indirect method as shown in Appendix C or the reference in 2.3.

For a given road operating condition, there are two intensity spectra obtained in this procedure, one at the tire leading edge, and the other at the trailing edge. The final results shall be the energy average of both intensity results.

5.2 Final Results. The final results should be presented in 1/3 octave bands sound power per Appendix G, which shows a simple empirical formula (a constant 3.2 dB shift) between sound intensity and sound power. To convert the 32 Hz narrow band signal into 1/3 octave bands, see Appendix F.

5.3 Tire Noise Articulation Index (TNAI). See Appendix E. The TNAI value can be used as a single number metric to quantify or rank the tire noise source.

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

7 Notes

7.1 Glossary.

Reference Tire: The Standard Reference Test Tire is commonly used throughout the industry to monitor differences in results and in some cases as a basis to adjust tire measurements to account for differences in test conditions, for example, different road conditions. Moreover, the results measured from the SRTT can be compared to historical results to assess whether the measurement is valid or unknown errors are involved.

Sound Intensity: Sound intensity is the acoustic energy flowing through a unit area in a sound field and, hence, is a vector quantity with an associated direction of propagation. Sound intensity is defined as watts per square meter, with reference value 1 pW/m² for Sound Intensity Level.

Sound Power: Sound power is acoustic energy radiated from a sound source in this case as tire noise. Its unit is watts, with reference value 1 pW used for Sound Power Level. Sound power can be calculated by measuring sound intensity and multiplying it by the intensity scanned area.

Sound Pressure: Sound pressure is the pressure variation that the human ear can detect. Sound pressure (as an effect) is a field response of the sound propagation in an environment where a sound source exists to radiate sound power (as a cause). Its unit is Pascals, or Newtons per square meter, with reference value 20×10^{-6} used for Sound Pressure Level.

7.2 Acronyms, Abbreviations, and Symbols.

Dyno	Dynamometer
ETRTO	European Tyre and Rim Technical Organization
PBNR	Power Based Noise Reduction
SPL	Sound Pressure Level
SRTT	Standard Reference Test Tire
TNAI	Tire Noise Articulation Index
TNC	Tire Noise Class
T&RA	Tire & Rim Association
VTs	Vehicle Technical Specification

8 Coding System

This standard shall be referenced in other documents, drawings, etc., as follows:

Test to GMW14239

9 Release and Revisions

This standard originated in August 2005, replacing GMN7079. It was first approved by the Global N&V Acoustics and Airborne Noise IDSR Harmonization Team in October 2005. It was first published in November 2005.

Issue	Publication Date	Description (Organization)
1	NOV 2005	Initial publication.
2	OCT 2006	Revised Section 2.2. (Global N&V Acoustics and Airborne Noise IDSR Harmonization Team)
3	MAR 2010	Merge GMW14235 into GMW14239. Revisions per comments from different regions. (Global N&V Acoustics and Airborne Noise IDSR Harmonization Team)

Appendix A: Alternative On Dyno Measurement Method

The following contents provide the specific instructions required by the dynamometer operation; otherwise refer to the on-road operation procedure in the main text.

A1 Procedure

A1.1 Preparation.

A1.1.1 Road Shell Installation. The specific road shell shall be properly installed in accordance with the test conditions. If necessary, a 2 to 3 mm thick layer of mastic damping or other damping absorption equivalents shall be placed on the roll beneath the shell cap joints to minimize abrupt joint impacts. Shells shall be uniformly bolted onto the roll with the specified installation torque (e.g., 47 N·m or 35 ft·lb as used on Milford Proving Ground Dyno) making sure that there is no abrupt vertical height difference between two adjacent shells. No loose bolts shall be allowed.

A1.1.2 Trailer Installation. The GM proprietary trailer as shown in Figures D5 and D6 shall be installed with its tires on the top of the shell surface and supported in the front by a ball-joint fixture that allows for free yaw rotation. A chain shall be utilized to secure the trailer to the fixture for additional safety. The fixture shall be placed at the centerline of the dynamometer rolls. The back side of the trailer shall be loosely secured to the dynamometer room ground with tie-down straps for safety (in a manner similar to a vehicle on the dynamometer).

A1.1.3 Removal of Stones. Any stones embedded inside the treads shall be removed for safety.

A1.1.4 Dynamometer Operation. Prior to normal dynamometer operation or equipment installation, the trailer should be centered on the rolls by rotating the dynamometer at low speed (3 to 6 km/h in a manner similar to vehicle centering on the dynamometer).

It is recommended to operate only one wheel of the dynamometer for tire noise measurement where possible.

A1.1.5 Tire Warm-up. Test tires shall be run to bring the tires up to their working temperature, equal to or more than 29 °C (85 °F), before the results are recorded. Warm-up usually takes 6 minutes at 80 km/h. A non-contact infrared temperature probe can be used to monitor the rolling tire temperature. If brand new tires are not broken in per 4.1.9, warm up the tire at least for 12 minutes at 80 km/h.

A1.1.6 Tire and Road Shell Cleanup. For brand new tires, see 4.1.8. The road shell surfaces shall be regularly cleaned up, to be free of any chemical or rubber residues from previous dynamometer tests. The operators can use 3M General Purpose Adhesive Cleaner or other equivalents (which will not damage the shell surface), soft brass-bristled brush, and paper towels to clean up.

A1.1.7 Microphone Positions.

A1.1.7.1 Near-Field Sound Intensity Measurement. Install the phase-matched microphone pair as shown in Figure D5. Refer to the main text or Figure D4 for positioning the microphone pair(s).

A1.1.7.2 Far-Field Sound Power Measurement. The sound power measurement standard, ANSI S12.35-1990, specifies a far-field sound power measurement method in a hemi-anechoic room. Several sound pressure responses are measured around the hemisphere that surrounds the rotating tire, as seen in Figures D6 and D7. In the test, a radius of 1.2 m is chosen to set the hemisphere to be in the far-field of the tire noise. The center of the hemisphere is located at the center of the tire contact patch. A tire radiates sound symmetrically around the tire central plane ($y = 0$ as shown in the coordinates in Figure D7) since its tread pattern is designed to be symmetric. Therefore, ten microphones as proposed in ANSI S12.35-1990 can be reduced to six microphones as proposed in Figure D7.

A1.1.7.2.1 Controllable Environment. A sheet of at least 35 mm thick open cell foam or equivalently high absorption materials shall be placed against the trailer side to avoid the reflection. The area adjacent to the test tire shall be free of any reflection objects except for the ground surface.

A1.2 Conditions.

A1.2.1 Environmental Conditions.

A1.2.1.1 Ambient Temperature. Test facility should have ambient temperatures between 18 and 29 °C (65 and 85 °F).

A1.2.1.2 Ambient Sound Pressure Level. The ambient sound pressure level at any measurement microphone location in the dynamometer chamber (with the chassis dynamometer energized but not rotating) shall not exceed 45 dB(A) in 1/3 octave bands from 200 to 10 kHz.

A1.3 Instructions.

A1.3.1 Tire/Dynamometer Speed. The tire/dynamometer speed shall be varied slowly during measurement to minimize dynamometer roll

periodic resonance orders and cap joint periodic impacts. The slow sweep will minimize these effects in the data average. The sweep speed starts at 5% lower and ends at 5% higher than the nominal speed. The speed shall run this range linearly up for 8 s and linearly down for another 8 s. For examples, for 60 km/h, sweep the dynamometer speed from 57 up to 63 km/h within 8 s, then down from 63 to 57 km/h in 8 s; for 80 km/h, sweep the dynamometer speed in the range 76 to 84 km/h.

A1.3.2 Near-Field Sound Intensity Measurement. See Appendices B and C. However, the time duration is recommended to be 32 s to capture several full sweep durations. With 50% overlap and resolution of 32 Hz, the average number should be 2048 averages for 32 s time duration. If only interested in 1/3 octave bands, one can use finer frequency resolutions.

A1.3.3 Far-Field Sound Power Measurement. Measure the six microphones SPLs at 1 Hz resolution and then convert the 1-Hz spectra into 1/3 octave bands spectra per the parameters set from GMW 8460.

A2 Data

A2.1 Calculations.

A2.1.1 Sound Intensity Calculation. See 5.1.

A2.1.2 Sound Power Calculation. The sound power calculation is based on ANSI S12.35-1990 far-field hemisphere measurement method. Use the following formula to calculate the total sound power level based on the six-microphone SPL dB values:

$$SWL = 10 \bullet \log_{10} \left[\sum_{i=1}^6 W_i \bullet 10^{(SPL_i / 10)} \right] + 10 \bullet \log_{10} (2\pi R^2) + 10 \bullet \log_{10} \left(\frac{400}{\rho c} \right)$$

Where:

SPL_i is the measured SPL (dB values) respectively for six microphones

R is the radius of the measurement hemisphere

ρ is the air density and

c is the speed of sound

Where:

$$R = 1.2 \text{ m}$$

$$\rho = 1.2 \text{ kg/m}^3 \text{ and}$$

$$c = 343 \text{ m/s}$$

W_i is the area weighted i -th surface contribution coefficient.

$$W_i = \begin{cases} 0.15 & i = 1 \\ 0.15 & i = 2 \\ 0.30 & i = 3 \\ 0.15 & i = 4 \\ 0.15 & i = 5 \\ 0.10 & i = 6 \end{cases}$$

Ensure that the microphone positions ($i = 1$ to 6) are the same as those shown in Figure D7.

A2.2 Data Correction from Dynamometer to Road. For a given tire and a running condition, one can calculate a sound intensity correction factor between road data acquired per on-road test (main text) and dynamometer data in this appendix. The correction factor can be stored and applied to future dynamometer measurement on the same type of tires for the same running condition to estimate road tire noise performance.

The data correction shall be done in the 1/3 octave bands.

Appendix B: Recommended Acquisition Parameters

Measure 32 Hz narrow band cross-power spectrum per the following acquisition parameters:

- Free run trigger
- 32 Hz resolution
- Maximum frequency: 6400 Hz or higher
- Hanning windows for all the channels
- 500 averages (minimum)
- 50% overlapping

Dynamic range: 72 dB (minimum)

Appendix C: Two-Microphone Cross-Spectral Intensity Method

The intensity can be calculated by the FFT based indirect method in JASA paper (see 2.3), that is, from the imaginary part of the cross-power sound pressure spectrum between the microphone pair.

$$I = -\frac{1}{\omega\rho\Delta r}\text{Im}(G_{12})$$

Where:

G_{12} is the cross-power spectra between two microphones, ω is the angular frequency, ρ is the air density, usually assumed to be 1.20 as a nominal value, varying from 1.15 to 1.27 (5~35°C), and Δr is the spacing between two microphones. The microphone spacing is recommended to be 16 mm in order to cover the frequency range of interest for tire airborne noise from 400 Hz to 6400 Hz.

Appendix D: Photos/Figures

Figure D1: Installation of Intensity Probe and Its Fixture in Test Trailer

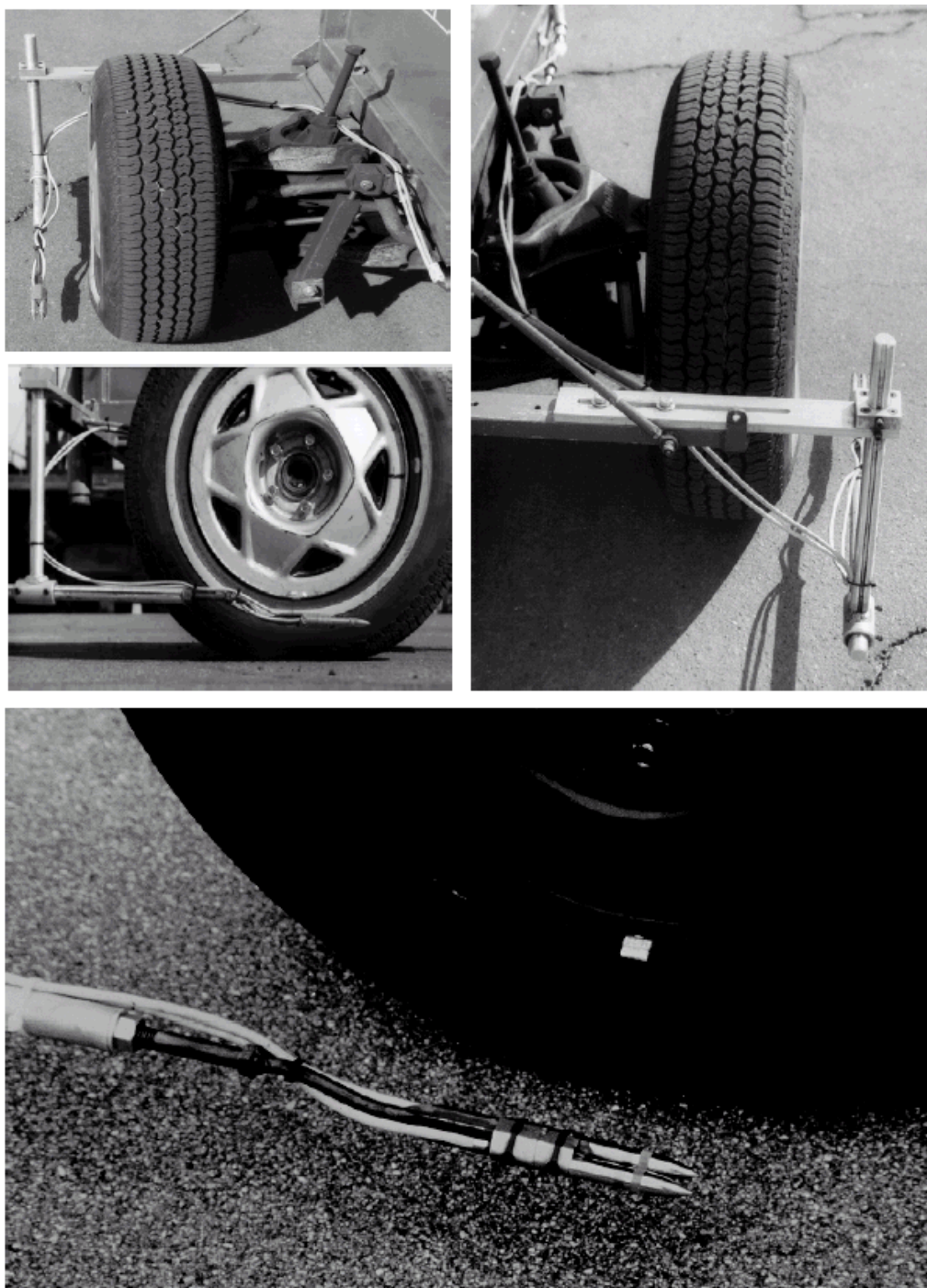


Figure D2: Test Trailer and Driving Vehicle on Test Road

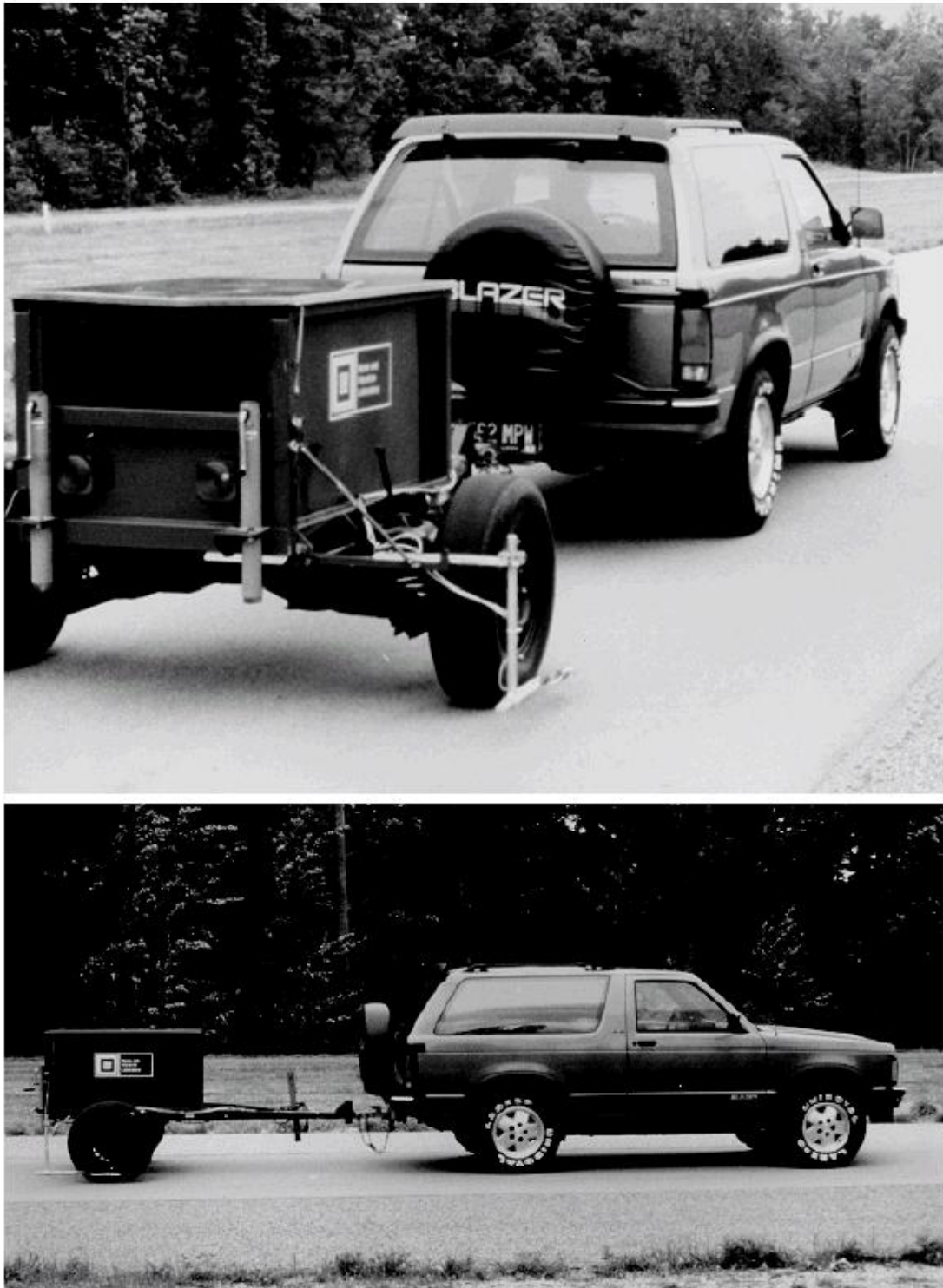


Figure D3: Installation of Intensity Probe and Its Fixture in Vehicle

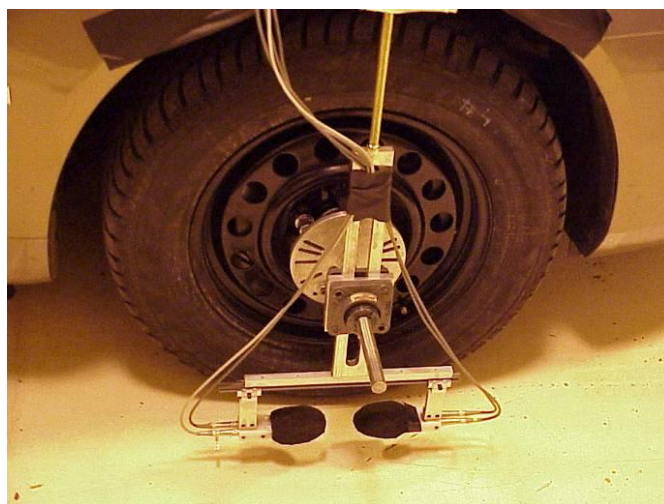


Figure D4: Schematic of Intensity Pair on Both Leading and Trailing Edges. See 4.3.3.3 for Positioning the Microphone Pair

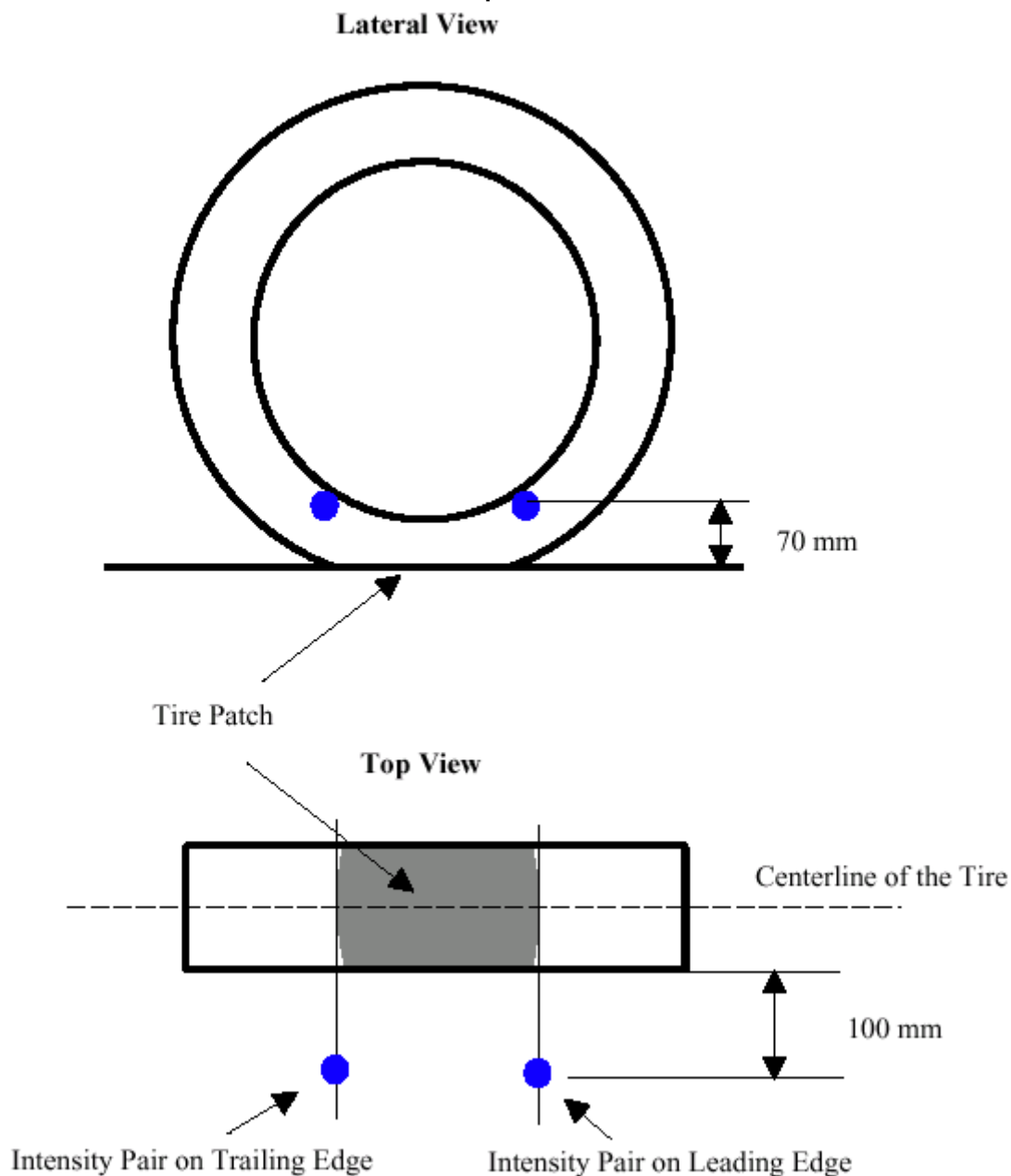


Figure D5: Fixture Setup with Intensity Probe Mounted

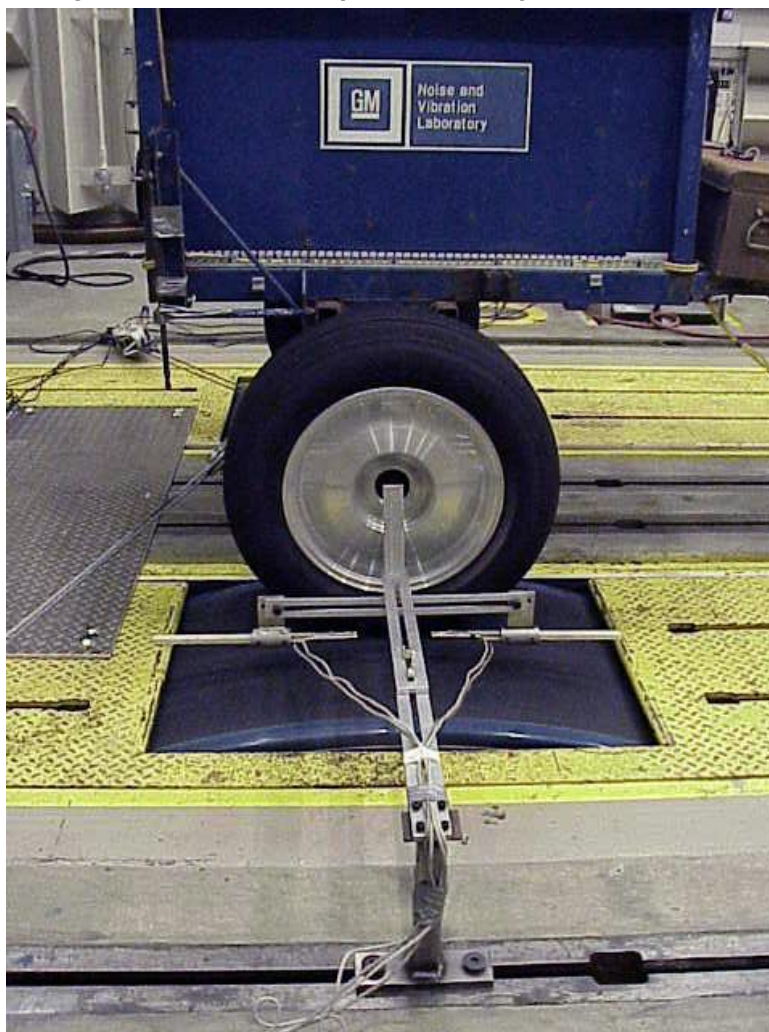


Figure D6: Trailer and Fixture Setup in Hemi-Anechoic Dynamometer Chamber

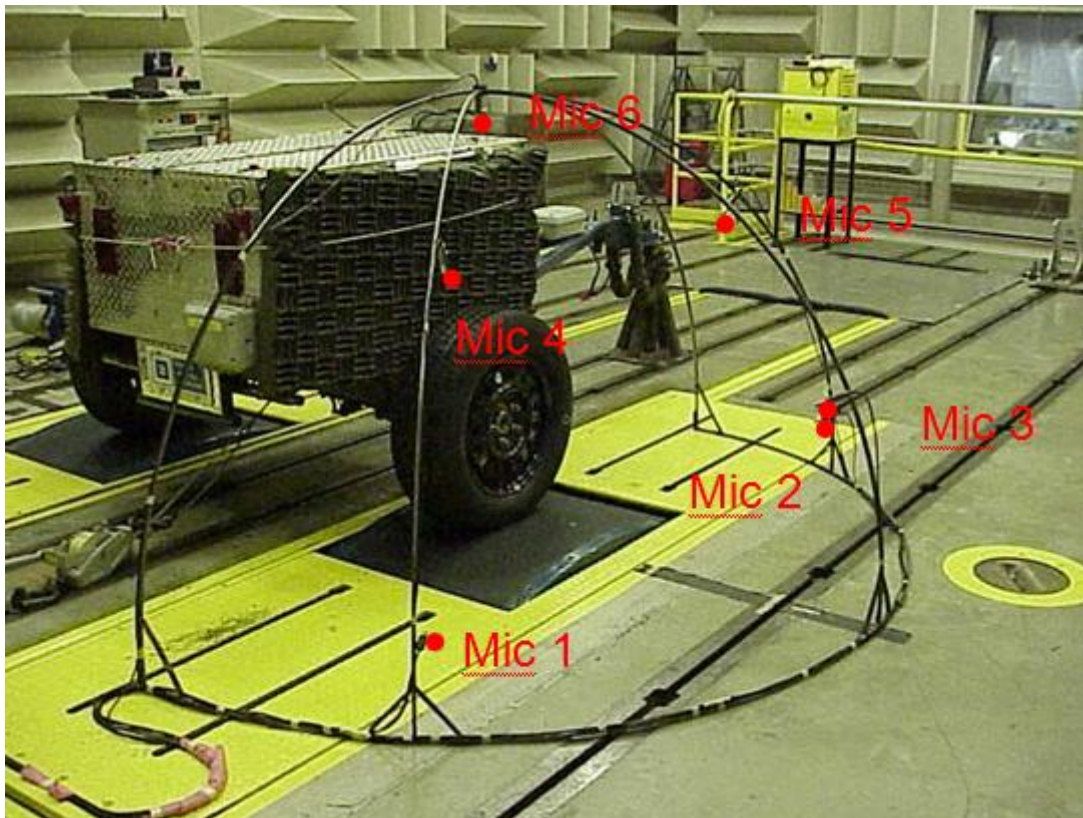
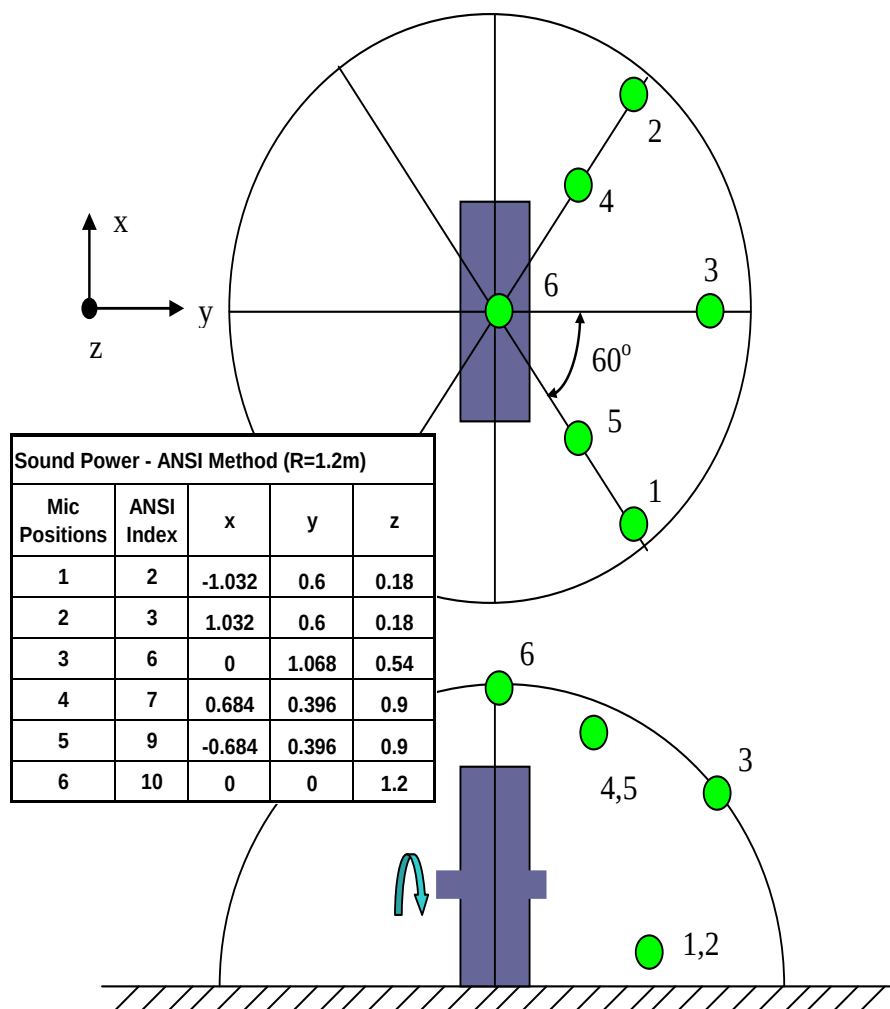


Figure D7: Schematic of Sound Power Measurement in Accordance with ANSI S12.35-1990



Appendix E: Tire Noise Articulation Index

A single number metric called Tire Noise Articulation Index (TNAI) has been developed to rank tire noise as a source. The TNAI values are calculated from the following steps:

1. Measure the tire noise sound power spectra (dB) in 1/3 octave bands from 400 Hz to 5000 Hz.
2. Subtract a sound power-based noise reduction (PBNR) in dB as shown below
3. Add 16.55 dB across the whole frequency range.
4. Calculate the Articulation Index (AI) value for the spectra from step 3) from 400 Hz to 5000 Hz.
5. Add 16.55 and 8.75 into the AI value.
 - 5.1) 16.55 is used to reverse Step 3.
 - 5.2) 8.75 compensate on the AI values from the frequencies other than 400 Hz to 5000 Hz.

Note that the TNAI value can be larger than 100. This index is empirically based.

The PBNR dB curve is given in the following table.

Hz	PBNR (dB)
400	33.99
500	37.40
630	41.20
800	44.51
1000	46.04
1250	48.11
1600	48.68
2000	50.56
2500	51.24
3150	53.12
4000	55.22
5000	57.86

Appendix F: Conversion of 32 Hz to 1/3 Octave Bands

The 32 Hz narrowband is preferable when there is a need to address tire sound quality issues. The user can use the recommended following formula to convert the 32 Hz narrow band signal into 1/3 octave bands.

$$I_i = \sqrt{B_i \sum_{j=1}^{n_i} (A_i^j)^2}$$

Where I_i is the i -th 1/3 octave band sound intensity amplitude. A_i^j is the j -th 32 Hz sound intensity spectrum amplitude within the i -th 1/3 octave band. n_i is the total 32 Hz spectrum lines within the i -th 1/3 octave band. B_i is the i -th 1/3 octave band correction constant which is defined as

$$B_i = \frac{f_i^U - f_i^L}{n_i \cdot 32}$$

This value compensates for the energy leaks due to not enough 32 Hz lines in some of 1/3 octave bands, particularly at low frequencies. f_i^U and f_i^L , respectively, are the upper band limit and lower band limit of the i -th 1/3 octave band, as shown in the following table.

Frequency (Hz)		
1/3 Octave Central Frequency	Lower Limit	Upper Limit
200	180	224
250	224	280
315	280	355
400	355	450
500	450	560
630	560	710
800	710	900
1000	900	1120
1250	1120	1400
1600	1400	1800
2000	1800	2240
2500	2240	2800
3150	2800	3550
4000	3550	4500
5000	4500	5600
6300	5600	7100
8000	7100	9000
10000	9000	11200

Appendix G: Sound Intensity to Sound Power

Once the 1/3 octave band sound intensity is obtained, one can calculate the sound power by using an empirical formula based on multiple measurements on 2 dynamometers.

$$SWL_i = SI_i - 3.2 \quad dB$$

where SWL_i is the i -th 1/3 octave band sound power level, SI_i is the i -th 1/3 octave band sound intensity level, and the constant 3.2 is related to the area as well as radiation directivity and units conversion.

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